

[Association between chronic urinary tract infection and primary biliary cirrhosis].

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The etiology of primary biliary cirrhosis (PBC) remains unclear, and at present, this disease is considered to be caused by the combined effect of genetic factors, infection, autoimmunity, and environmental factors. Since infection is the major cause for PBC, scholars have been focusing on the association between chronic microbial infection. Studies have shown that *Escherichia coli* is the most common bacterium for urinary tract infection (UTI), and recurrent UTI has been confirmed to be a risk factor for the development and progression of autoimmune liver diseases and is closely associated with PBC. This article investigates the association between UTI and PBC and possible mechanisms.

Tackling antimicrobial resistance in lower urinary tract infections: treatment options.

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Urinary tract infections (UTIs) are among the most common infectious diseases occurring in either the community or healthcare settings. A wide variety of bacteria are responsible for causing UTIs, however extra-intestinal pathogenic *E. coli* or ExPEC) remains the most common etiological agent. Since 2000, resistance to antibiotics emerged globally among ExPEC and is causing delays in appropriate therapy with subsequent increased morbidity and mortality. The aim of this review article is to provide an overview on the definitions, etiology, treatment guidelines (including agents for infections due to antimicrobial resistant bacteria) of lower UTIs and to highlight recent aspects on antimicrobial resistance of ExPEC. Expert commentary: For patients with acute uncomplicated lower UTIs, nitrofurantoin, trimethoprim-sulfamethoxazole, fosfomycin or pivmecillinam should be prescribed for a 1-5 day course depending on the agent used. Single-dose fosfomycin is an excellent option for uncomplicated lower UTIs and has had similar clinical and/or bacteriological efficacy for 3- or 7-day regimens for alternate agents (i.e., ciprofloxacin, norfloxacin, cotrimoxazole or nitrofurantoin).

Revisiting the Frequency and Antimicrobial Resistance Patterns of Bacteria Implicated in Community Urinary Tract Infections.

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Urinary tract infections (UTIs) are one of the most common infectious diseases at the community level. The continue misuse of antimicrobials is leading to an increase in bacterial resistance, which is a worldwide problem. The objective of this work was to study the incidence and pattern of antimicrobial resistance of the main bacteria responsible for UTI in the community of central and northern Portugal, and establish an appropriate empirical treatment. The urine samples were collected in Avelab—Laboratório Médico de Análises Clínicas over a period of 5 years (2015–2019). The urine cultures were classified as positive when bacterial growth was equal to or higher than 10⁵ CFU/mL, and only for these cases, an antimicrobial susceptibility test was performed. Of the 106,019 samples analyzed, 15,439 had a urinary infection. Urinary infections were more frequent in females (79.6%) than in males (20.4%), affecting more elderly patients (56.9%). *Escherichia coli* (70.1%) was the most frequent uropathogen, followed by *Klebsiella pneumoniae* (8.9%). The bacteria responsible for UTI varied according to the patient's sex, with the greatest differences being observed for *Enterococcus faecalis* and *Pseudomonas*

aeruginosa, these being more prevalent in men. In general, there was a growth in bacterial resistance as the age of the patients increased. The resistance of bacteria in male patients was, in most cases, statistically different (Chi-Square test, $p < 0.05$) from that observed for bacteria isolated from female patients, showing, in general, higher resistance in male patients. Although *E. coli* was the most responsible uropathogen for UTI, it was among the bacteria most susceptible to antibiotics. The isolates of *K. pneumoniae*, *Proteus vulgaris* and *Enterobacter* showed high resistance to the tested antimicrobials. The most common multidrug-resistant (MDR) bacteria implicated in UTI were *K. pneumoniae* (40.4%) and *P. aeruginosa* (34.7%), but *E. coli*, the most responsible bacteria for UTI, showed a MDR of 23.3%. When we compared our results with the results from 10 years ago for the same region, in general, an increase in bacterial resistance was observed. The results of this study confirmed that urinary tract infections are a very common illness, caused frequently by resistant uropathogens, for which the antibiotic resistance profile has varied over a short time, even within a specific region. This indicates that periodically monitoring the microbial resistance of each region is essential in order to select the best empirical antibiotic therapy against these infections, and prevent or decrease the resistance among uropathogenic strains.

Acquisition of a pathogenicity island in an *Escherichia coli* clinical isolate causing febrile urinary tract infection.

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Urinary tract infections (UTI) are less common in men than in women, and uropathogenic *Escherichia coli* (UPEC) is the most frequent etiological agent. Recurrent UTI in men have often been reported as a relapse with the same strain as the index infection. The persistence of the same *E. coli* strain within the urinary tract has often been explained by a prostatic focus. The aim of this study was to determine whether recurrence was associated with relapse or reinfection and the possible effect of treatment on the content of virulence factors of the isolates causing these infections. Thirty *E. coli* isolates were collected from 15 patients with a febrile UTI having a bacteriological recurrence during long-term follow-up. These isolates were analyzed by pulsed-field gel electrophoresis (PFGE) and virulence profiling in order to determine whether they constituted relapse or reinfection. Five recurrences were categorized as relapse and nine as reinfections. The results obtained showed that the horizontal transfer of virulence factors contained in a pathogenicity island had occurred in one isolate. This event is possible in vivo and allows bacteria to become more virulent and, perhaps, cause greater damage. The acquisition of virulence genes by horizontal gene transfer is an ongoing process of evolution that continuously leads to new bacterial pathotypes.

[Bacterial urinary tract infection and subclinical bacteriuria in the dog: a current review].

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Bacterial urinary tract infection (UTI) is a common clinical presentation in dogs and a frequent reason for the prescription of antimicrobial drugs. UTI refers to adherence, multiplication and persistence of an infectious agent within the urogenital system. This causes an associated inflammatory response as well as the pertaining clinical signs. Depending on the site of infection, UTI's may be classified as bacterial

cystitis, prostatitis or pyelonephritis. In contrast, subclinical bacteriuria (SB) is defined as the presence of a significant number of bacteria in the urine of an individual not showing clinical signs referable to UTI. UTI's typically occur as a consequence of ascending pathogen migration from the host's own fecal or distal urogenital microbial flora. The most commonly isolated pathogen in cases of UTI and SB is *Escherichia coli*. The diagnosis is based on clinical signs and the results of urine examination and culture. The recently revised guidelines of the International Society for Companion Animal Infectious Diseases provide detailed recommendations for diagnosis and management of different forms of bacterial UTI's in dogs. Adherence to treatment guidelines will improve treatment success and is imperative in avoiding further deterioration of the antimicrobial resistance situation.

Urinary tract infections among under-five children with congenital anomalies of kidney and urinary tract in hospitalized patients in Addis Ababa.

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Congenital anomalies of the kidney and urinary tract (CAKUT) and urinary tract infections (UTIs) are significant clinical concerns in children. Children who have UTIs are also at risk of developing long-term complications, notably kidney disease and scarring UTIs. The purpose of this study was to assess the prevalence of urinary tract infection and its determinants among under-five children with congenital anomalies of kidney and urinary tract in Addis Ababa. A health institution-based retrospective cross-sectional study was employed on 389 randomly selected medical records of children with CAKUT, out of a total of 850 cases. The children were admitted to Tikur Anbesa and Menelik II Hospitals between January 1, 2018, and December 31, 2022. The study hospitals were purposively selected as they were the only referrals in the nation for CAKUT cases. Sociodemographic, clinical, and laboratory characteristics were captured using a data extraction sheet created in the Kobo Toolbox and transferred to SPSS for analysis. Binary logistic regression analysis was used to evaluate the associations of variables with a p-value of ≤ 0.05 and a confidence interval of 95%. The majority of the study participants (336 [86.4%]) were males. A total of 194 (49.8%) children were older than 25 months, with a median age of 23 (95% CI: 22-27) months, and 117 (30.1%) (95% CI = 25.4-35.2) participants were positive for urinary tract infection (UTI). Forty were culture-positive out of 103 urine specimens tested, where, *Escherichia coli* (60%) was the most frequently isolated bacteria followed by *Klebsiella pneumonia* (20%), *Enterobacter* spp, (12.5%), and *Staphylococci* spp. (7.5%). Factors associated with UTI among CAKUT children were found to be being female (AOR = 4.3, 95% CI = (1.4-13.07), children having Posterior Ureteral Valve (PUV) (AOR = 4.8, 95% CI = (2.32-9.93), duration of hospitalization (> 4 days) (AOR = 3.08, 95% CI = (1.52-6.24), and children having fever (AOR = 5.8, 95% CI = (2.81-11.97). In this study, almost one-third of the children developed UTIs, and female sex, longer hospitalization, PUV, and fever were identified as independent risk factors of UTIs. This emphasizes special attention should be given to high-risk children.

The respiratory chain of *Klebsiella aerogenes* in urine-like conditions: critical roles of NDH-2 and bd-terminal oxidases.

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Klebsiella aerogenes is an opportunistic nosocomial bacterial pathogen that commonly causes urinary tract infections. Over the past decades, *K. aerogenes* strains have acquired resistance to common

antibiotics that has led to the rise of multidrug-resistant and even pandrug-resistant strains. Infections produced by these strains are nearly impossible to treat, which makes *K. aerogenes* a global priority to develop new antibiotics and there is an urgent need to identify targets to treat infections against this pathogen. However, very little is known about the metabolism and metabolic adaptations of this bacterium in infection sites. In this work, we investigated the respiratory metabolism of *K. aerogenes* in conditions that resemble human urine, allowing us to identify novel targets for antibiotic development. Here we describe that, unlike other gram-negative pathogens, *K. aerogenes* utilizes the type-2 NADH dehydrogenase (NDH-2) as the main entry point for electrons in the respiratory chain in all growth conditions evaluated. Additionally, in urine-like media, the aerobic metabolism as a whole is upregulated, with significant increases in succinate and lactate dehydrogenase activity. Moreover, our data show that the bd-I type oxidoreductases are the main terminal oxidases of this microorganism. Our findings support an initial identification of NDH-2 and bd-I oxidase as attractive targets for the development of new drugs against *K. aerogenes* as they are not found in human hosts.

Human and avian extraintestinal pathogenic *Escherichia coli*: infections, zoonotic risks, and antibiotic resistance trends.

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Extraintestinal pathogenic *Escherichia coli* (ExPEC) constitutes ongoing health concerns for women, newborns, elderly, and immunocompromised individuals due to increased numbers of urinary tract infections (UTIs), newborn meningitis, abdominal sepsis, and septicemia. *E. coli* remains the leading cause of UTIs, with recent investigations reporting the emergence of *E. coli* as the predominant cause of nosocomial and neonatal sepsis infections. This shift from the traditional Gram-positive bacterial causes of nosocomial and neonatal sepsis infections could be attributed to the use of intrapartum chemoprophylaxis against Gram-positive bacteria and the appearance of antibiotic (ATB) resistance in *E. coli*. While ExPEC strains cause significant healthcare concerns, these bacteria also infect chickens and cause the poultry industry economic losses due to costs of containment, mortality, and disposal of carcasses. To circumvent ExPEC-related costs, ATBs are commonly used in the poultry industry to prevent/treat microbial infections and promote growth and performance. In an unfortunate linkage, chicken products are suspected to be a source of foodborne ExPEC infections and ATB resistance in humans. Therefore, the emergence of multidrug resistance (MDR) (resistance to three or more classes of antimicrobial agents) among avian *E. coli* has created major economic and health concerns, affecting both human healthcare and poultry industries. Increased numbers of immunocompromised individuals, including the elderly, coupled with MDR among ExPEC strains, will continue to challenge the treatment of ExPEC infections and likely lead to increased treatment costs. With ongoing complications due to emerging ATB resistance, novel treatment strategies are necessary to control ExPEC infections. Recognizing and treating the zoonotic risk posed by ExPEC would greatly enhance food safety and positively impact human health.

Antibiotic Resistance, RAPD- PCR Typing of Multiple Drug Resistant Strains of *Escherichia Coli* From Urinary Tract Infection (UTI).

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Global spreading of multidrug resistant strains of *Escherichia coli* is responsible for Urinary Tract Infection (UTI) which is a major health problem in of concern. Among the gram negative bacteria, the major contributors for UTI belongs to the family Enterobacteriaceae, which includes *E. coli*, *Klebsiella*, *Citrobacter* and *Proteus*. However, *E. coli* accounts for the major cause of Urinary tract infections (UTIs) and accounts for 75% to 90% of UTI isolates. The main aim of this study is to analyse the phylogenetic grouping of clinical isolates of UTI *E. coli*. In this study nearly 58 *E. coli* strains were isolated and confirmed through microbiological, biochemical characterization. The urine samples were collected from outpatients having symptoms of UTI, irrespective of age and sex in Tamil Nadu, India. The isolates were subjected to analyse for ESBL and AmpC β -lactamase production. To understand its genetic correlation, molecular typing was carried out using RAPD-PCR method. Here we noted phenotypically twenty seven isolates were positive for ESBL and seven for AmpC β -lactamase production. However, among the ESBL isolates higher sensitivity was noted for Nitrofurantoin and Cefoxitin. It is worth to note that the prevalence of UTIs was more common among female and elderly male. Phylogenetic grouping revealed the presence of 24 isolates belonged to B2 group followed by 19 isolates to group A, eight isolates to group B1 and Seven isolates to group D. Phenotypically most of the strains were positive for ESBL and showed high sensitivity for Nitrofurantoin and cefoxitin.

[Urinary tract infections. Views on microbiological diagnosis and resistance determination].

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Urinary tract infection is a common problem affecting every fifth woman during her life. Although the clinical diagnosis is easy, and is based mainly on symptoms and microscopy of urine sediment, the microbiological diagnosis is often difficult. *Escherichia coli* is the main etiologic agent, causing urinary tract infection in approximately 80% of the cases. Adherent, uropathogenic *E. coli* strains are associated with pyelonephritis and relapse. *Staphylococcus saprophyticus* is a frequent cause of urinary tract infections in fertile women. The urethral syndrome, caused by several agents, is an important factor in differential diagnosis. Microscopy is the cornerstone of all diagnostics of urinary tract infections. A urine culture that is taken, handled, transported and interpreted satisfactorily can provide valuable information with respect to diagnosis, treatment, prophylaxis and epidemiology.